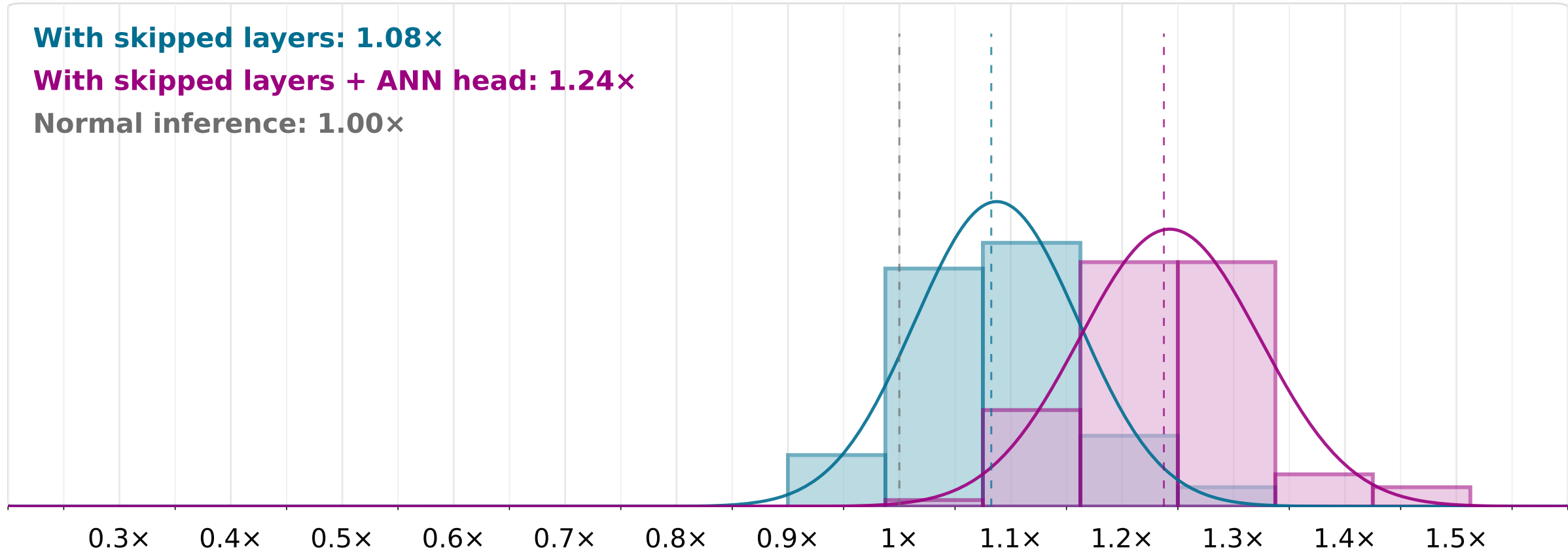


# Self-speculation per-token speedup

## Llama 3.2 1B Instruct



### Setup

Prompt set = **python diverse**  
Gap = **(1, 1)**  
Block size = **2**  
Layers = **16**  
LM vocab = **128,256**  
LM params = **1.24B**  
Head params = **262.7M**  
Head portion = **21.25%**

### Runtime

GPU = **NVIDIA L4**  
Backend = **sdpa**  
Model dtype = **bfloat16**  
Bridge dtype = **bfloat16**  
Speed internals = **no**  
Profile prompts = **15**  
Warmup prompts = **5**  
Measured prompts = **100**

### Results

Peak mem normal = **2.34 GiB**  
Mean speedup (w/o ANNH) = **1.08×**  
Acceptance rate (w/o ANNH) = **41.2%**  
Exact match (w/o ANNH) = **47.0%**  
Peak mem self (w/o ANNH) = **2.35 GiB**  
Mean speedup (w ANNH) = **1.24×**  
Acceptance rate (w ANNH) = **40.2%**  
Exact match (w ANNH) = **47.0%**  
Peak mem self (w ANNH) = **2.39 GiB**  
Tokens gen = **s 19,978 / n 20,000**

### Profile

Drafter split (w/o ANNH) = **B 43.9% / H 50.0% / O 6.1%**  
Verifier/normal (w/o ANNH) = **1.07×**  
Drafter/normal (w/o ANNH) = **33.6%**  
Drafter split (w ANNH) = **B 75.3% / H 14.3% / O 10.4%**  
Verifier/normal (w ANNH) = **1.07×**  
Drafter/normal (w ANNH) = **19.4%**  
ANNH accuracy = **97.6%**  
ANNH head speedup = **6.03×**  
ANNH index = **8016 clusters; top-k 100**